

Port Memory

Design a Single-port BRAM (depth 2048, width 16)

//Verilog code

//Design single port BRAM (depth 2048 width 16), 16x2048, how many BRAM it is consuming?

`timescale 1ns / 1ps

```
module singleport_BRAM_16x2048(  
    input wire clk,  
    input wire we,  
    input wire [10:0] addr,  
    input wire [15:0] din,  
    output reg [15:0] dout);  
  
    (* ram_style = "block" *) reg [15:0] mem [0:2047];  
  
    always @(posedge clk) begin  
        if (we)  
            mem[addr] <= din;  
        dout <= mem[addr];  
    end  
endmodule
```

2. Memory

Site Type	Used	Fixed	Available	Util%
Block RAM Tile	1	0	140	0.71
RAMB36/FIFO*	1	0	140	0.71
RAMB36E1 only	1			
RAMB18	0	0	280	0.00

RTL analysis schematic:

